

Declaration on the Use of Generative AI Tools

SABD Project 2 – Real-Time Analysis of Flight Delays with Apache Flink

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Purpose

In accordance with the Project 2 specification (*“Utilizzo di strumenti di Intelligenza Artificiale Generativa”*), this document declares the use of generative AI tools during the development of the project. The aim is to provide a transparent, honest, and complete account of where AI was used, where it was not, and what was reviewed and validated by the author.

Tools used

- **Anthropic Claude** (model: `claude-opus-4-7`), accessed via the Claude.ai desktop application in Cowork mode. Used in conversational sessions throughout the project.

No other generative AI tools (Copilot, ChatGPT, Gemini, etc.) were used during the development of this project.

Where generative AI was used

- **Brainstorming.** Discussing architectural alternatives for stream ingestion (Kafka vs. socket vs. file source) and their trade-offs given the constraints of a single-student team.
- **Code generation.** Initial drafts of the following artefacts were generated by Claude and then reviewed, executed, and corrected by the author:
 - `docker/Dockerfile.flink`, `docker/docker-compose.yml`.
 - `feeder/feeder.py`, `feeder/Dockerfile`.
 - `src/preprocess.py`, `src/utils.py`.
 - `src/query1.py`.
 - Four iterative versions of `src/query2.py` (PyFlink attempts) and the final `src/query2_batch.py` (pandas implementation).
 - `benchmarks/run_q1_benchmarks.sh`.
- **Debugging.** Interpreting runtime errors and stack traces during development (PEP 668 `pip` restriction, FileSink directory permissions in bind-mounted volumes, PyFlink 1.20 absence of `socket_text_stream` in its Python API, Beam Python worker memory limit and the resulting out-of-memory failures of the Q2 PyFlink iterations).
- **Report writing.** Drafting the LaTeX source of the report (`Report/report.tex`) and of this declaration. Section structure and final wording were reviewed and edited by the author.

Where generative AI was NOT used

- **Design decisions.** The choice of architecture (Python TCP feeder + Flink + post-processing), of the acceleration factor ($\alpha = 60,000$), of the watermark strategy (bounded out-of-orderness 60 s), and of the hybrid stream/batch split for Q2 were made by the author after weighing the trade-offs.
- **Diagnosis of Q2 throughput ceilings.** The conclusion that PyFlink could not sustain Q2 under the deployment constraints (single-host laptop, 1.7 GB TaskManager budget, ~ 360 keyed states) was reached by the author from repeated empirical observation across the four iterations, and motivated the move to the hybrid design.
- **Experimental setup.** The choice of metrics (total replay time, throughput, output-equality verification across parallelism values), the experimental protocol, and the interpretation of the results were the author's.
- **Final review.** Every generated artefact was read, executed, and validated before being included in the project. Generated code was modified where it did not pass review. Generated report text was edited to match the author's knowledge and voice.

Accountability

The author is able to discuss and justify every design decision, implementation detail, and experimental result of the project during the oral defence, independently of the involvement of the AI tool. The author retains full responsibility for the correctness and appropriateness of the delivered artefacts.